METR — MODEL EVALUATION & THREAT RESEARCH

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every 7 Months

50% task-completion time horizon doubling every ~7 months

Thomas Kwa, Ben West, Joel Becker et al.

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arxiv.org/abs/2503.14499

METHODOLOGY

A New Metric: 50% Task-Completion Time Horizon

The task duration at which an AI model achieves 50% success — an intuitive, human-grounded measure of autonomous capability

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?" Choices: "run.sh", "run.txt", "run.py", "run.md"
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

1 **Assemble Task Suite**

170 diverse software/research tasks spanning 1 second to 8 hours

2 **Measure Performance**

Time skilled humans + measure AI agent success rates on each task

3 **Fit Logistic Model**

Estimate task duration at which each AI agent achieves 50% success rate

TASK SUITE

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

Three complementary task suites cover the full spectrum of autonomous software work, grounded in 800+ human baselines (2,529 hours)

Source	Tasks	Human Time Range	Description	Example Tasks
SWAA Software Atomic Actions	66	1 sec – 30 sec	Short single-step software tasks	find_shell_script (3s): "Which file is a shell script?"
HCAST Human-Calibrated AI Software Tasks	97	1 min – 30 hrs	Diverse software tasks across many domains	wikipedia_research (1 min) oxdna_simple (9 min) — fix bug in molecular dynamics simulation munge_data (56 min) — write Python to transform JSON data
RE-Bench Research Engineering Benchmark	7	8 hrs (all tasks)	Difficult ML research engineering tasks	cuda_backtesting (8 hrs): Implement custom CUDA kernels for 30x speedup of backtesting tool

170 Total Tasks

800+ Human Baselines

2,529 Hours of Human Work

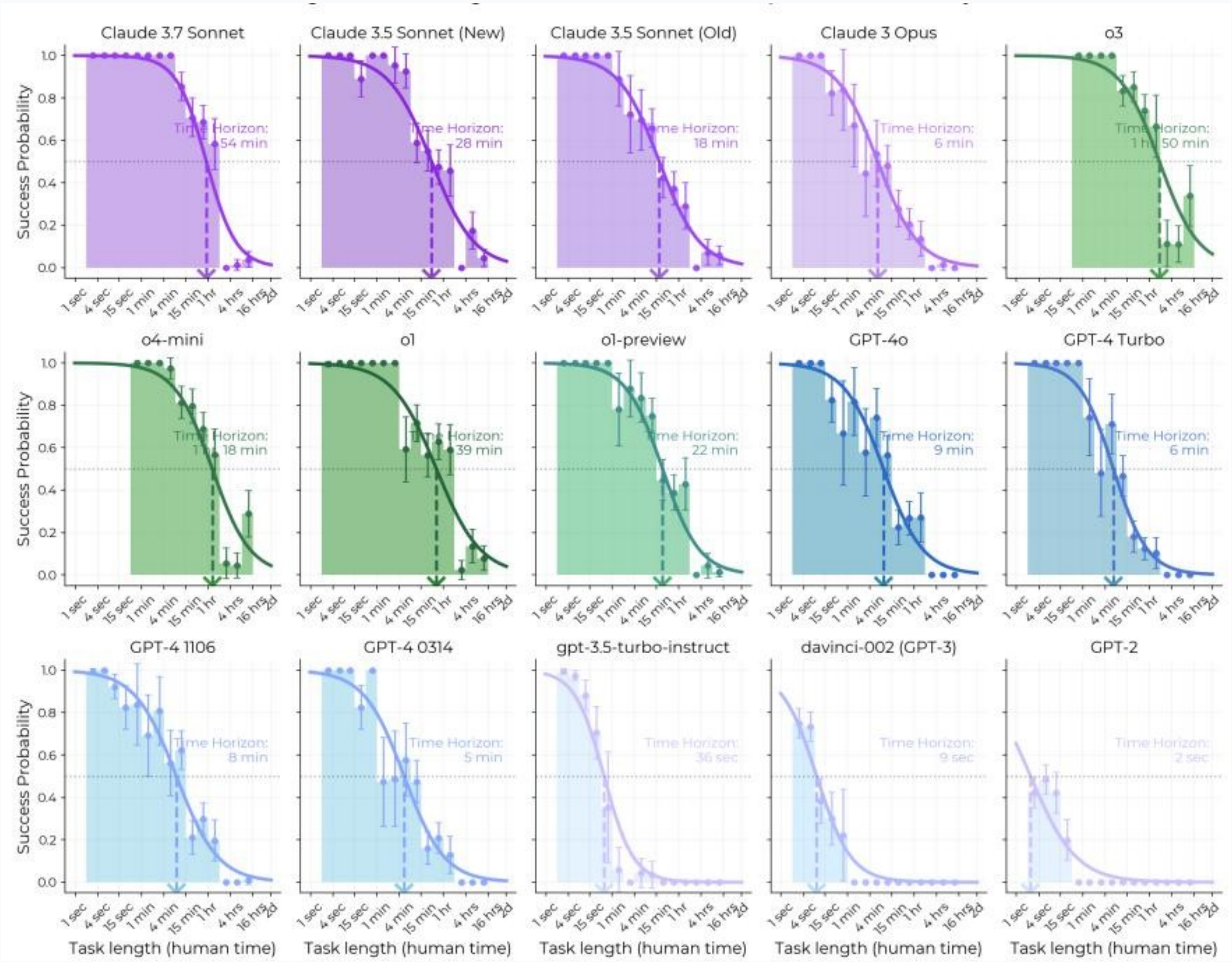
3s – 8h Task Duration Range

Source: Kwa et al. (2025). Task durations measured by skilled human professionals.

EXPONENTIAL TREND

AI Task Horizon: 2 Seconds (GPT-2) to ~2 Hours (o3) in 6 Years

50% time horizon has grown exponentially since 2019 with a doubling time of ~207 days (~7 months) and $R^2 = 0.97$



DOUBLING TIME

~207 days (~7 months)

FIT QUALITY

$R^2 = 0.97$

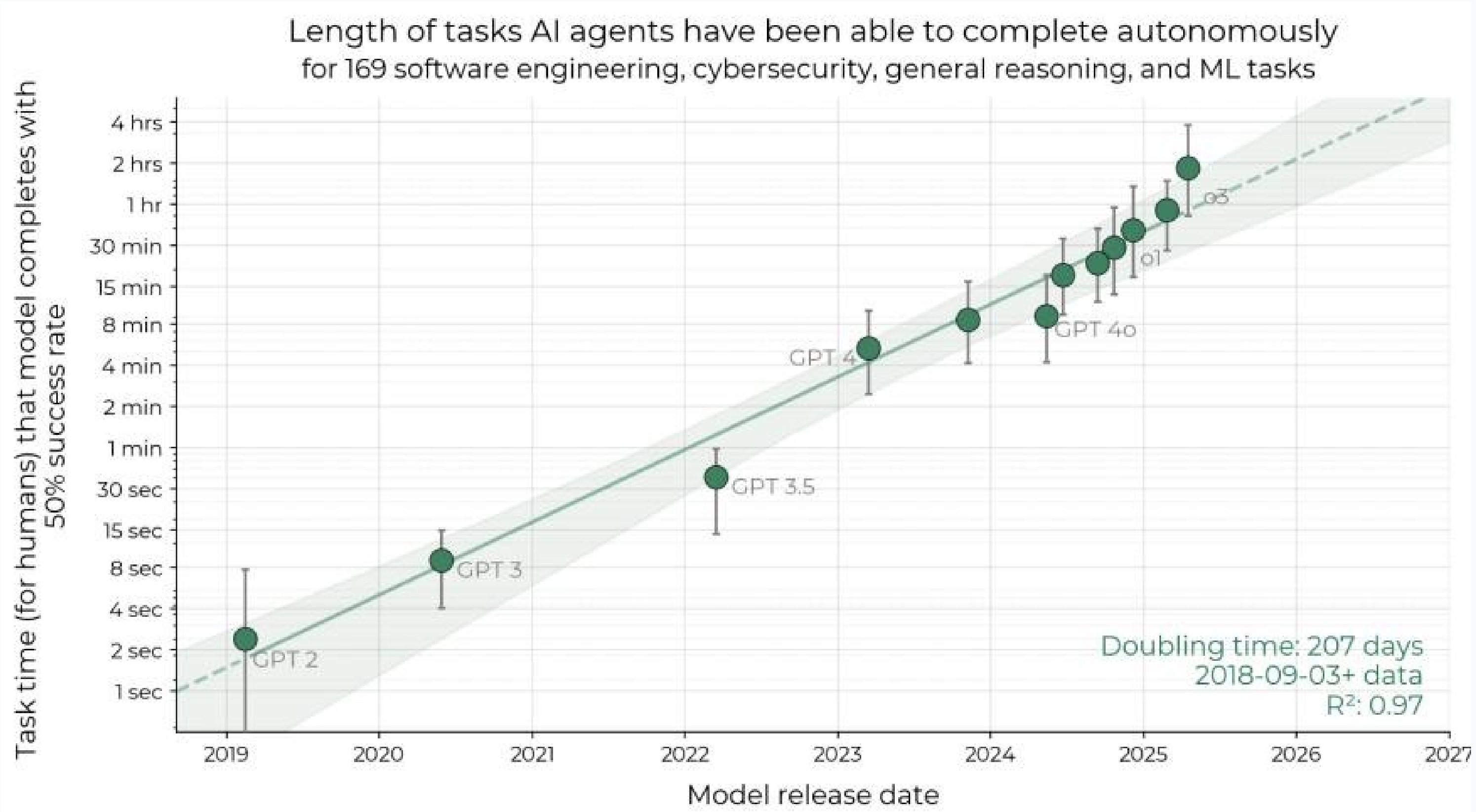
KEY MODEL MILESTONES

- **GPT-2 (2019)** ~2 seconds
- **GPT-4 (2023)** ~5 minutes
- **Claude 3.5 Sonnet (2024)** ~28 min
- **o3 (2025)** **~1h 50m**

2023–2025 growth rate ~20% faster than overall trend

Success Probability Drops Sharply as Task Length Increases for All Models

Frontier models push the success curve rightward — but all models show steep decline beyond their time horizon



KEY INSIGHT

Each model's success rate declines with task length. Frontier models shift the curve rightward.

GPT-2 (2019)
~2 sec
50% time horizon

Claude 3.7 Sonnet (2025)
~54 min
50% time horizon

o3 (2025)
~1h 50m
50% time horizon

80% time horizon (tasks completed 80% of the time) shows similar trend but horizons are ~5x shorter

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

Four capability improvements explain the exponential growth — but performance remains notably lower on unstructured tasks

Improved Logical Reasoning

Models can decompose complex multi-step problems into coherent solution plans, enabling longer chains of autonomous work.

Better Tool Use

Effective use of shells, APIs, file systems, and development tools — the scaffolding for extended autonomous software tasks.

Greater Reliability & Self-Awareness

Models better recognize when they are stuck or producing incorrect output, reducing compounding errors over long task horizons.

Ability to Adapt to Mistakes

When encountering errors or unexpected results, frontier models can diagnose and correct course — critical for longer tasks.

Key Limitation

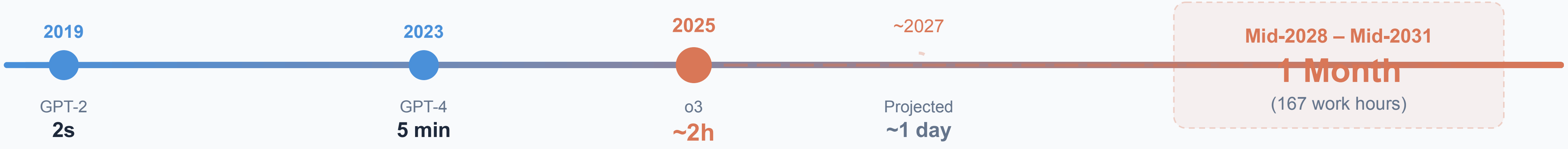
Performance is notably lower on less structured, "messier" tasks. External validity remains an open question — the task suite does not perfectly represent the average segment of real-world intellectual labor.

FORWARD EXTRAPOLATION

Extrapolation Predicts Month-Long Task Automation by 2028–2031

Naive extrapolation: AI reaches 1-month time horizon (167 work hours) between mid-2028 and mid-2031, automating many month-long projects

Doubling every ~7 months



2023–2025 growth is ~20% faster than 2019–2025 rate

Caveats & Uncertainties

External validity uncertain
Tasks may not perfectly represent the distribution of real-world intellectual labor

Trend may change
Exponential growth rarely continues indefinitely — saturation, bottlenecks, or architecture shifts could alter trajectory

Supplementary experiments found little evidence that trends are slower on more realistic tasks

KEY TAKEAWAYS

Exponential Autonomous Capability Growth Demands Proactive Safety Governance

1. **Robust, Intuitive Metric**
50% time horizon provides a single number for tracking AI autonomous capability growth.
$R^2 = 0.97$ over 6 years

2. **Rapid Doubling with Possible Acceleration**
Doubling every ~7 months. The 2023–2025 rate is ~20% faster than the overall 2019–2025 trend.
~7 mo doubling time

3. **Current Frontier: ~2 Hours of Human-Equivalent Work**
o3 can autonomously complete tasks that take skilled humans nearly 2 hours — up from 2 seconds for GPT-2.
~2 hrs o3 frontier

4. **External Validity Remains an Open Question**
Tasks may not perfectly represent real-world intellectual labor distributions — but supplementary experiments found little evidence that trends are slower on more realistic tasks.

5. **Key Limitations: Unstructured Tasks & Uncertain Generalization**
Performance is notably lower on "messier" tasks. Generalization to real-world job distributions uncertain. Safety governance frameworks must account for both the trend and its uncertainty bounds.

Source: Kwa et al. (2025) — METR